Curriculum Vitae

Personal Statement

An adaptable and detail-oriented Software and Embedded Systems Engineer with a strong background in data pipelines, system optimization, and technical component delivery. Proven track record of developing robust software testing frameworks, writing Linux kernel drivers, and building predictive neural networks in Python.

A pragmatic, high-agency engineer eager to transition into a mission-driven role within the Ministry of Justice AI Unit, applying strong problem-solving skills to build and deploy practical, user-focused AI solutions across complex operational environments.

Technical Skills

- **Programming Languages**: C, C++, Python (PyTorch, NumPy, Pandas)
- **Embedded Systems**: I2C, SPI, CAN protocols; RISCV, Arduino microcontrollers
- **Software Development**: Linux systems, thread management, CMake, Jenkins, Bash
- **Circuit Design**: PCB analogue/digital design (Altium, KiCad); PCB milling and fabrication
- **Simulation Tools**: MATLAB, SolidWorks, Ansys, COMSOL for thermal-mechanical simulation
- **Prototyping & Testing**: Wire-bonding, soldering, laser milling; TFT probing and analysis

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Experience

TouchNetix, Fareham – Embedded Engineer

September 2025 – Present

- Developing a touchscreen controller driver for mainline Linux Kernel (LKML). This involves navigating the strict upstreaming process, adhering to kernel coding standards, and managing patch submissions.
- Collaborating directly with the hardware team during chip bring-up to develop self-test routines. Identified and leveraged underutilized analog front end resources, cutting latency from 2s to 50ms and improving factory throughput.
- Investigated and resolved firmware bugs and I2C/SPI timing issues using logic analysers and oscilloscopes to ensure robust communication between the controller and peripherals.

Accelercomm, Southampton – 5G Software Engineer

October 2023 – September 2025

- Leading company-wide system compliance to the product specification and 3GPP 5G NR standards.
- Developed a new testing framework to validate 3GPP 5G physical channels, generating 5G NR slot/PDU scenarios using MATLAB.
- Implemented DPDK-based hardware accelerator drivers, enhancing system efficiency.
- Acquired in-depth knowledge of standards including 3GPP, O-RAN, nFAPI, and eCPRI.
- Collaborating on code-reviews to ensure code quality standards and consistent approach across the team.

Formula Student – Instrumentation Lead

September 2022 – July 2023

- Designed and tested analogue processing PCBs for the control of the EV power system.
- Led the development of a CAN-based telemetry network, integrating sensors using I2C/SPI with EV systems for real-time data collection.

- Designed and tested PCBs fitted with STM32 microcontrollers configured with RTOS thread management, optimized for multi-priority task handling.

CSA Catapult, Newport – Advanced Packaging Intern

July 2021 – July 2022

- Improved wire bonding and die attachment processes for micro-scale MMIC and VCSEL dies.
- Conducted thermo-mechanical simulations, reducing thermal stress in semiconductor designs.
- Collaborated with industry partners to prototype innovative RF and power semiconductor applications.

Additional Roles

- **Training Captain, University Athletics Club** (2020–2021): Designed tailored training plans for diverse skill levels.
- **Volunteer, Foodbank & Refugee Community Kitchen** (2019–2020): Coordinated logistics and service delivery under time-sensitive conditions.

Education

MEng Electrical and Electronic Engineering
2 Scholarships – overall 2:1

Part 1 – 69.2 %
Modules include: Engineering Management, Electrical Circuits and Power Systems, Electromagnetics & RF Engineering, Digital Electronics, Power Electronics and Applications, Integrated Circuit Engineering, Optoelectronics and Wireless Communications & Digital Signal Processing.

Part 2 – 70.4 %

Modules include: Advanced Embedded Systems, Terahertz Science and Technology, Microengineering & High Frequency Electronics.

A-Levels: Maths (A*), Further Maths (A*), Physics (A*), Chemistry (A)
GCSEs: 10 subjects including Maths, Physics, and Chemistry (A*/A/B)

Projects & Interests

- **AI Development**: Built and trained convolutional neural networks (CNNs) for predictive image classification tasks using PyTorch and Kaggle datasets.
- **Hobbyist Projects**: Explored microcontroller communication protocols and designed small-scale PCB systems.
- **Scientific Research Enthusiast**: Regularly follows advancements in physics, cosmology, and AI via sources such as Sabine Hossenfelder and PBS Space Time.
- **Personal Pursuits:** A keen runner, ranked in the Top 1000 in the UK (Half Marathon 1hr10, 5k 15:30). Self-taught classical piano player.